

Pham Viet Thanh

[Thanhbo209](#) | [Pham Thanh](#) | [Portfolio](#) | thanhagar123@gmail.com | [0396200543](#)

Summary

Software engineering student with hands-on experience building full-stack applications and AI-integrated tools across backend APIs, database design, and frontend interfaces. Comfortable working across the stack from REST API design and data modeling to React-based UIs with growing depth in AI engineering fundamentals. Looking to contribute as a full-stack while building toward a long-term focus in AI engineering.

Education

HCMC University of Foreign Languages and Information Technology
Bachelor of Software Engineering

Nov 2024 – Jul 2027

Relevant Coursework

Data Structures & Algorithms	Object-Oriented Programming
Database Systems	Software Testing & Quality Assurance
Project Management	Web/App Development
Design Patterns	Business Analyst

Skills

- Languages:** JavaScript/TypeScript, Python, HTML/CSS
- Frontend Frameworks:** React, Next.js, Tailwind CSS, Vite
- Backend Development:** Node.js, Express, FastAPI, RESTful APIs, JWT Authentication
- AI & Data:** NumPy, Pandas, AI API Integration, AI-assisted Development
- Databases (SQL/NoSQL):** PostgreSQL, MongoDB, Prisma, Redis
- Tools & Practices:** Git, GitHub, Docker, Postman, CI/CD

Experience

Backend AI Engineering Intern
FlyRank AI

Jun 2026 – Present
Remote

- Building backend services and REST APIs using TypeScript, Node.js, PostgreSQL, and modern software engineering practices to support AI-powered applications.
- Developing production-ready backend features including authentication, authorization, asynchronous background jobs, and containerized development workflows with Docker.
- Integrating AI APIs into backend applications and implementing document ingestion pipelines to support AI-driven workflows and data processing.
- Designing interactive workflow orchestration interfaces with React Flow while collaborating through Git-based development, debugging, testing, and code reviews.
- Applying AI engineering principles and AI-assisted development practices throughout a production-oriented capstone project, while earning Anthropic Claude certifications.

IT Labeling / Product Annotation Contributor
Acacy Co., Ltd.

May – June 2026
On-site

- Annotated product images in CVAT with bounding boxes and category labels, maintaining consistent object boundary precision across images for downstream computer vision model training
- Filtered and curated raw product images for AI training suitability, screening out low-quality, ambiguous, or mislabeled samples before dataset handoff
- Applied inter-label consistency standards across annotation batches, reducing rework flagged during QA review

Certifications

• Google AI Essentials – Google

Jul 2026

Understood prompt engineering strategies and AI-driven workplace productivity.

• Claude Code 101 – Anthropic

Jul 2026

Hands-on experience with Claude Code, agentic workflows, prompt engineering, and context management for AI-assisted development.

• TOEIC Listening & Reading – Score: 870

Jun 2026

Administered by IIG Vietnam (Official ETS Representative). Verified advanced professional English proficiency.

Projects

LeetCode DSA Tracker

Sep 2026

TypeScript, Next.js, Prisma, Supabase, Chrome MV3, Gemini API, Vitest

[GitHub](#) | [Live Site](#)

- Engineered a full-stack algorithmic knowledge platform with a Chrome Manifest V3 extension that auto-detects the active deployment origin, captures LeetCode submissions by polling the official GraphQL submission-details API across a background service worker and content-script bridge, and syncs code to the backend through shared authenticated session cookies.
- Integrated Google Gemini to audit algorithmic time/space complexity and synthesize structured knowledge drafts for a 3-tier Knowledge Vault (Approach → Solution → Code), persisting validated AI output as typed Prisma JSON fields.
- Designed a resilient dual-model AI failover system (Gemini 3.6 ↔ 2.5) with concurrency-safe atomic state transitions and quota-category-aware cooldowns (per-day vs. short-window rate limits), backed by an audit-logged admin dashboard tracking live model status and switch history.
- Resolved production connection-pool exhaustion (15-connection cap under concurrent serverless invocations) by migrating to Supabase's Supavisor transaction pooler with Prisma's adapter-pg, then layered ISR (on-demand revalidatePath from mutation routes) and Router Cache stale-time tuning to eliminate redundant database queries on repeat page loads.
- Built public profile playbooks with in-memory multi-criteria filtering (difficulty, topic, search) synced to the URL via the History API to avoid server round-trips, memoized filtering/rendering with paginated result sets to minimize recomputation, plus session-based route protection via a Next.js proxy layer securing private API routes and dashboard pages while preserving public profile access.
- Maintained 274 unit and integration tests across 32 Vitest suites enforcing data isolation and filter-logic correctness, with CI-enforced strict TypeScript linting achieving zero any types across the entire codebase.

Retail Product Detection

May 2026 – Present

Python, YOLOv8, CVAT, OpenCV, Pandas, NumPy

[GitHub](#)

- Built a one-class retail product detection pipeline using YOLOv8, converting shelf annotations to YOLO format with bounding-box validation, label visualization, training, and inference export with per-image product counts
- Designed a CVAT human-in-the-loop labeling workflow that separated model pre-labels from corrected ground truth, improving precision from 0.249 to 0.635, mAP50 from 0.090 to 0.408, and detecting 117/120 products on a dense shelf image at confidence above 0.50
- Experimenting with augmentation configurations (mosaic, HSV shifts, horizontal flip) and confidence/IoU threshold tuning to push mAP50 beyond current 0.408 baseline; evaluating multi-class expansion to include shelf labels as a second detection target

AI Resume Analyzer

Jan - Feb 2026

Javascript, React Router v7, Tailwind CSS, Puter.js, Zustand

[GitHub](#) | [Live Site](#)

- Architected a serverless full-stack app on Puter.js's cloud platform, using its hosted Claude Sonnet integration for resume scoring and multi-turn AI feedback without managing a backend or API keys directly
- Implemented client-side request deduplication and response caching via Zustand, cutting redundant Sonnet API calls by over 60%

- Built session-based authentication through Puter.js's native auth layer and integrated its cloud file storage with global sharing, letting users generate a shareable public link to their analyzed resume and scoring breakdown